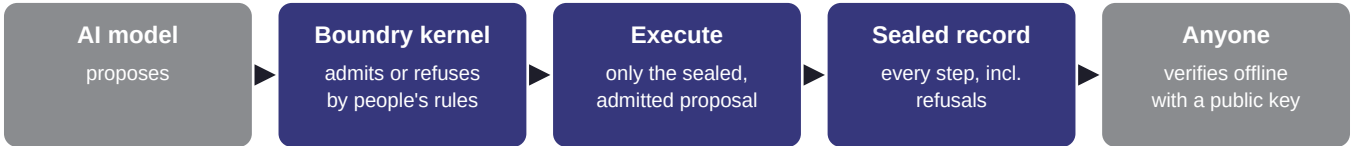


A working control for the actions of AI agents

Deterministic, model-free admission of every action — and a record anyone can check

Most approaches to AI safety look for the answer inside the model — training it, prompting it, fencing it in. But the harms we fear from AI agents are harms of **action**: money moved, records changed, messages sent, systems operated. The safety of an action does not have to live inside the model. Boundry lets the model propose freely, and puts execution in a deterministic engine that only carries out what rules set by people allow. It decides before, and records after, in a record anyone can check.

How it works — a deterministic guardrail at the action layer



The model's only output is a *proposal*. A kernel containing no model checks it against fixed rules — known kind, permitted operations, the requester's tenant and no other — and admits it or refuses it with a stated reason. Only an admitted proposal runs, and it runs from the sealed proposal's bytes, never from the model's live output. Every stage, including every refusal, is sealed into a signed, hash-chained record. In layered-guardrail terms, Boundry is a fail-closed guardrail at the tool-execution and action stage — deterministic rather than model-based, and leaving a verifiable record of every decision it makes.

Mapped to the controls in the AI Safety Institute report *Risks and controls for multi-agent systems* (9 August 2026)

Control the report describes	What Boundry does
Persistent agent identity; actions logged against it for traceability	Every record is attributed to the principal that invoked the AI — never to the model — and every decision, admitted or refused, is sealed against that identity.
Agent identity with principal binding	Requests are scoped to the requester's tenant. The author of each record is the invoking principal. Gating by recorded authority is designed, not yet built.
Rollback to a last-known-good state via checkpoints	The record is append-only and signed checkpoints commit to it; state is rebuilt from the record, and errors are corrected by new entries rather than edits, so history is never lost.
Limiting what an agent can do, including under manipulation	The model has no direct path to action. A prompt-injected instruction can change what the model proposes, but cannot make a proposal pass that the declared rules do not permit.
Oversight and assurance	Records are verified offline by an independent verifier that shares no code with the kernel, using only a public key — without access to the operator's systems. A public verifier and sample records are available.

Where it stands — stated plainly

Element	Status
Sealing and signed record layer	In live use in a bookkeeping application and, since October 2026, in a financial reconciliation application. In the reconciliation application, the application checks daily that its records match the chain, and holds approvals on any mismatch. Each deployment signs with its own key; independent time-stamping and a rebuild-from-empty on live data are not yet in place.
Full governing core (rule-based admission and refusal, execution from sealed bytes)	Demonstrated on a test bench with synthetic data: requests sealed and verified end to end; state rebuilt from the record and compared identical; one altered byte refused and the record named; two independent verifiers in agreement; the software rebuilt byte-for-byte on a second machine.
Independent time-stamping (RFC 3161) of checkpoints	Built, not yet enabled.
Governing core in production use	A later stage.

What it does not do

Boundry does not judge whether rules are good — it applies them faithfully and makes them attributable. It governs only actions routed through it; protection is as complete as the rule that Boundry is the sole path to action. If a model proposes something wrong but permitted, it is admitted, so consequential actions should still have human confirmation. Records are tamper-evident, not tamper-proof: an edit is not prevented, but cannot pass unnoticed.

Why it matters for Australia

Regulators have made clear that existing obligations apply in full to AI, so organisations increasingly need evidence of what an AI agent did and that it was authorised — not just assurances. Boundry produces that evidence as a by-product of operating. Nothing in it depends on the author being a model: the same separation of declared intent, deterministic admission and independent verification applies to any system where work can be stated before it is done. It was built in Australia, by a practitioner in tax and accounting, and is running in that sector today.

What I would value

Your view on where this sits relative to current guardrail research, and on the preprint at github.com/Boundryos/research; and, if it is of interest, a short demonstration.

“The model proposes. A deterministic engine disposes. Every step is sealed — and anyone can check it.”

Ben Mueller · Boundry · verify@boundry.tech · verify.boundry.tech
Try the verifier and sample records at verify.boundry.tech. Patent pending (Australian provisional applications).